# The Kinetics of Electron Transfer through Ferrocene-Terminated Alkanethiol Monolayers on Gold

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The kinetics of electron transfer between a substrate gold electrode and a self-assembled monolayer formed from $CH_3(CH_2)_{n-1}SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_nSH$ were studied as a function of $n$, the number of methylenes in the alkyl chain tethering the ferrocene moiety to the electrode, using the indirect laser-induced temperature jump method (ILIT). For $5 \leq n \leq 9$ the standard electron-transfer rate constants vary according to $k_{\Gamma,s,n=0} \exp[-\beta_n n]$ where $k_{\Gamma,s,n=0}$ is the (extrapolated) rate constant for the electron transfer at $n = 0$. At $T = 25$ °C, $k_{\Gamma,s,n=0} \simeq 6 \times 10^8$ s$^{-1}$ and $\beta_n = 1.21 \pm 0.05$. The ILIT method allows rates to be measured that are too fast to be measured by conventional chronoamperometry at a macroelectrode, which is limited to rate constants of $\lesssim 10^4$ s$^{-1}$. Using a Marcus formalism, the reorganization energy, $\lambda$, for the electron-transfer process at a given $n$ was determined from the slope of an Arrhenius plot over the temperature range 15−55 °C. Values of $\lambda$ determined from Arrhenius slopes for $n = 8$ and 9 using ILIT are in reasonable agreement with the value of $\lambda$ previously deduced from the potential dependence of the rate constant for $n = 16$. For $n \leq 8$ the ILIT data show a decrease in the value of $\lambda$ as $n$ decreases; the decrease is too large to be explained by image charge interactions. The data also suggest that $|V_n|$, the $n$-dependent electronic coupling term, does not increase as rapidly as expected with decreasing $n$ for $n < 8$.

## Introduction

Meaningful characterization of the dependence of a heterogeneous electron-transfer rate constant upon the distance between the electrode and the redox moieties demands that the distance be well defined. Li and Weaver[1] studied the irreversible reduction of cobalt(III) complexes attached to gold and mercury surfaces by thioalkylcarboxylate ligands. They found that the rate of this reaction decreased exponentially with alkyl chain length on gold surfaces. However, the structure of the interface in the Li and Weaver system was not well characterized. The ideal arrangement would be to attach the redox moiety to the electrode surface with a structurally well-defined molecular tether, analogous to the approach used in studies of intramolecular electron transfer in solution.[2] Using such an arrangement, Chidsey[3] examined the rate of electron transfer between a gold electrode and surface-attached ferrocene which is part of a self-assembled monolayer formed from a ferrocene-terminated alkanethiol, $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_nSH$, and an alkanethiol diluent, $CH_3(CH_2)_{n-1}SH$, where $n = 16$ ($n$ is the number of methylenes in the alkyl chain of the ferrocene compound). The films had been well characterized, and previous studies[4,5] of these mixed monolayer films had shown that they are electrochemically well-behaved as long as the mole fraction of the ferrocene compound is less than ~0.25. Because $n = 16$, the relatively slow rate of electron transfer could be measured by chronoamperometry, a conventional electrochemical technique, even at high overpotentials.[6] The rate of electron transfer across homologous monolayer systems for $n \geq 7$ are also slow enough to be characterized using chronoamperometry.[7] Those studies and earlier measurements on ruthenium pentamine pyridine-terminated thiol monolayers made by Finklea and coworkers[8,9] demonstrate that the standard electron transfer rate constant, $k_{\Gamma,s}$,[10] varies exponentially with $n$:

$$k_{\Gamma,s} = k_{\Gamma,s,n=0} e^{-\beta_n n} \quad (1)$$

where $\beta_n$ is the exponential decay coefficient, and $k_{\Gamma,s,n=0}$ is the extrapolated value of the rate constant for $n = 0$. Alternatively eq 1 can be expressed in terms of the distance, $r$, between the electrode and the redox moiety:

$$k_{\Gamma,s} = k_{\Gamma,s,r=0} e^{-\beta_r r} \quad (2)$$

where $\beta_r$ is the exponential decay coefficient. If the only source of $r$ dependence in $k_{\Gamma,s}$ is the electronic coupling between donor and acceptor sites, $\beta_r$ may be taken as the decay coefficient for electronic tunneling commonly referred to as $\beta$.[11] However, it is possible that the reorganization energy, $\lambda$, is also $r$ dependent. To the extent that the interfacial electron transfer involves image charges in the metal electrode, the value of $\lambda$ may decrease with $r$, effecting an increase in $k_{\Gamma,s}$ as discussed later.[12,13] Such effects must be accounted for before the purely electronic decay coefficient, $\beta$, can be extracted from the distance dependence of the rate constant, $k_{\Gamma,s}$. A related procedure is commonly used in the analysis of homogeneous electron transfer data.[14]

Because of the large separation between the electrode and the ferrocene redox moiety, the electron transfer rates were slow enough when $n = 16$ to be studied using conventional chronoamperometry[6] at large overpotentials, $E - E°$. The behavior of heterogeneous electron transfer in the high-overpotential regime can be successfully described by Marcus theory;[12] the Butler−Volmer description of electrode kinetics is inadequate in the high-overpotential regime.[3] When $E - E° \ll -\lambda/e$ (where $\lambda$ is the reorganization energy, and $e$ is

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the electronic charge) the value of the electron transfer rate constant controlling reduction, $k_{\Gamma,f}$, becomes independent of potential; when $E - E^\circ \gg \lambda/e$ the value of the electron-transfer rate constant controlling oxidation, $k_{\Gamma,b}$, becomes independent of potential. This is the electrochemical analogue of the now familiar inversion region for highly exergonic homogeneous electron-transfer reactions.[2,12] We can rewrite Chidsey's[3] formulation of the Marcus relationship

$$k_{\Gamma,f} = \frac{P_n}{\lambda^{*1/2}} \int_{-\infty}^{\infty} \frac{\exp(-[x - (\lambda^* + E^*)]^2/4\lambda^*)}{1 + e^x} dx \quad (3)$$

$$k_{\Gamma,b} = \frac{P_n}{\lambda^{*1/2}} \int_{-\infty}^{\infty} \frac{\exp(-[x - (\lambda^* - E^*)]^2/4\lambda^*)}{1 + e^x} dx \quad (4)$$

where $\lambda^* = \lambda/k_BT$, $E^* = e(E - E^\circ)/k_BT$, $k_B$ is the Boltzmann constant, and

$$P_n = 2\pi^{3/2}|V_n|^2\varrho_m/h \quad (5)$$

where $|V_n|$ is the electronic coupling factor (dependent upon $n$), $\varrho_m$ is the density of electronic states in the metal, $h$ is Planck's constant, and where $\lambda$ may have some $n$ dependence. The basic assumptions underlying eqs 3–5 are that the density of states in the metal and the electronic coupling are independent of energy. When $E = E^\circ$ (i.e., $E^* = 0$) one obtains the standard rate constant

$$k_{\Gamma,s} = k_{\Gamma,s,n=0}e^{-\beta_n n} = \frac{P_n}{\lambda^{*1/2}} \int_{-\infty}^{\infty} \frac{\exp(-[x - \lambda^*]^2/4\lambda^*)}{1 + e^x} dx \quad (6)$$

In addition to Finklea and co-workers[8,9] a number of other workers have been studying the distance dependence of heterogeneous electron transfer. Creager and co-workers[15,16] have used cyclic voltammetry and chronoamperometry to extract rate constants and reorganization energies for surface attached ferrocene moieties (for *N*-(15-mercaptopentadecyl)ferrocenecarboxamide,[15] $CpFeCp(CH_2)_8SH$,[16] and $CpFeCpCO_2(CH_2)_8SH$).[16] Miller and co-workers,[17–20] using a different, but related approach, controlled the electron transfer reaction distance by interspersing a self-assembled monolayer of electroinactive $\omega$-hydroxythiols between the electrode and the plane of closest approach of the redox moieties in the solution phase. For this type of system, great care must be taken to ensure that the monolayer is free of pinholes or other types of defects[21] which greatly facilitate electron transfer. Even with a perfect film, the electron-transfer rate is partially diffusion controlled and this complicates the analysis. When the redox moieties themselves are attached irreversibly to the surface, there is no need to have redox species in solution and the effects of pinholes and defects are minimized.

Of particular interest in this paper is the characterization of the distance dependence of the electron-transfer rate constant for an alkanethiol-tethered ferrocene system when $n \leq 8$, and the rate constant is too fast to measure using conventional electrochemical methods (e.g., chronoamperometry). We want to learn whether or not the exponential increase in the rate constant observed for $n \geq 8$[7] continues as $n$ is decreased. We are also interested in exploring the behavior of the reorganization energy, $\lambda$, as the redox moieties get close to the metal electrode. Liu and Newton[22] have proposed a three-layer dielectric continuum model for computing the solvent reorganization energy for film-modified electrodes. (The inner sphere contribution for the ferrocenium + e ⇌ ferrocene interconversion is less than or equal to 0.03 eV.[23]) For the case of an ideal metal electrode (i.e., a perfect conductor), their theory predicts that

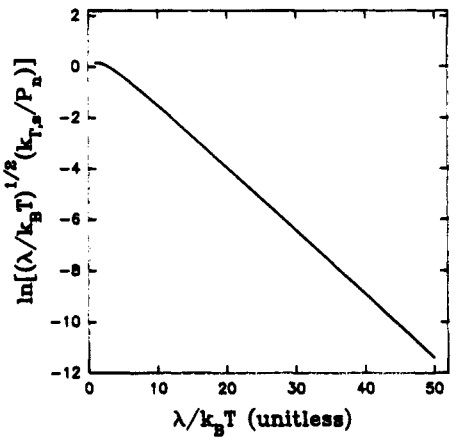



**Figure 1.** Semilogarithmic plot of $(\lambda/k_BT)^{1/2}(k_{\Gamma,s}/P_n)$ versus $(\lambda/k_BT)$ where $P_n$ is defined in eq 5.

the value of $\lambda$ should decrease with decreasing values of $n$ because of interactions between the charge on the redox moiety and its image in the metal electrode. In the present paper we use the indirect laser-induced temperature jump (ILIT) method[24,25] to characterize the rates of electron transfer between a gold electrode and a surface-attached ferrocenium/ferrocene redox couple for $5 \leq n \leq 9$. The electron-transfer rates for this system for $n < 7$ are too fast to be measured using chronoamperometry, thus excluding the use of Chidsey's direct method for evaluating $\lambda$ by chronoamperometry with large potential steps $|E - E^\circ|$ (i.e., comparable to $\lambda/e$).[3] The ILIT method is a small perturbation method which works best when the redox system is poised in the vicinity of $E = E^\circ$. The perturbation is much too small to evaluate $\lambda$ by driving the system into the high-overpotential regime. It is possible, however, to obtain a value of $\lambda$ from the measured temperature dependence of the electron-transfer rate constant. A plot of the integral in eq 6 versus the dimensionless quantity $\lambda^*$ ($=\lambda/k_BT$) is shown in Figure 1. The limiting slope of this plot for large values of $\lambda^*$ is $-1/4$. Over the temperature range used here (15–55 °C) and over the range values of $\lambda$ found in this work (0.7–0.96 eV), the plot in Figure 1 is essentially linear with an average slope of $-1/4.03$. Implicit in this analysis is the assumption that $\lambda$ is the sole contributor to the activation energy for electron transfer (this is not always the case; see Forster and Faulkner[26]). Assuming, further, that $\lambda$ is independent of temperature, we can write:

$$\lambda = -4.03 \frac{d \ln[k_{\Gamma,s}/T^{1/2}]}{d[1/T]} \quad (7)$$

where $d \ln[k_{\Gamma,s}/T^{1/2}]/d[1/T]$ is simply the slope of a plot of experimental values of $\ln[k_{\Gamma,s}/T^{1/2}]$ vs $1/T$ (an Arrhenius plot). We use this expression to determine $\lambda$ from ILIT data.

## Experimental Section

The ILIT apparatus, cell, and experimental techniques have all been described previously.[24,25] The objective of the method is to effect a rapid (<5 ns) and small (<5 K) change in the temperature at an electrode/electrolyte interface. This is accomplished by using a pulsed Nd:YAG laser (full width at half-maximum = 8 ns) to heat the "back" side of a thin (~1 $\mu$m) metal film electrode whose "front" side (with attached monolayer) contacts the electrolyte. Some of the laser energy is absorbed by the metal film and (in < 1 ps) degraded to heat which rapidly diffuses across the electrode raising the temperature of the electrode/electrolyte interface. The increase in temperature effects a change in the open-circuit potential (the experimentally measured quantity). The relaxation of the open

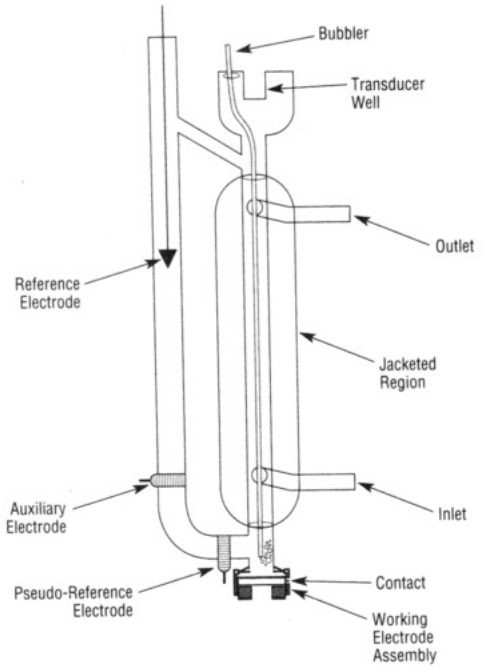



**Figure 2.** Diagram of the jacketed cell used in the ILIT experiments which measured the temperature dependence of the electron-transfer rate constant. See Figure 3 in ref 25 for a complete description of the working electrode assembly.

circuit potential is a function of the dissipation of the thermal perturbation and of the rate of electron transfer between the electrode and the redox moieties.

When ILIT experiments were carried out over a range of temperatures, a jacketed cell (see Figure 2) was used. The temperature of the electrolyte solution in the jacketed region of this cell is controlled by water flowing through the jacket from a NESLAB RTE-9DD Endocal refrigerated circulating bath. The gas bubbler (see Figure 2) circulates the electrolyte solution so that the temperature of the working electrode approximates that of the solution in the jacketed region of the cell. The temperatures of both the solution in the vicinity of the working electrode and the solution in the jacketed region of the cell were measured and calibrated (versus the temperature of the circulating bath) using an immersible thermocouple. The temperature of the solution in the jacketed region was 55.7 °C when the temperature in the vicinity of the working electrode was 55.0 °C. As expected, the temperature differential is zero when operating at ambient temperature. The lowest temperature used was about 15 °C. Some experiments were carried out using a virtually identical but unjacketed cell[25] operating at room temperature (29.5 ± 1.5 °C).

To monitor the size of the temperature jump at the working electrode/electrolyte interface, a pressure transducer (Panametrics, Waltham, MA) is placed in the well (containing purified water) at the top of the jacketed cell pictured in Figure 2. The temperature jump induces a pressure pulse, the size of which is directly proportional to the interfacial temperature change. This pressure pulse moves up through the electrolyte solution, is transmitted through the glass which forms the bottom of the transducer well, moves through the water in the well, and is then measured by the transducer whose output is amplified and recorded along with the ILIT signal.[25] The transducer response was calibrated using high (and equal) concentrations of ferrocyanide and ferricyanide as previously described.[25]

The gold film electrodes used in this work were vapor deposited over a titanium underlayer on quartz substrates (1 in. diameter disks, made of Homosil quartz by Heraeus Amersil). The titanium underlayer was ~500 Å thick, and these electrodes

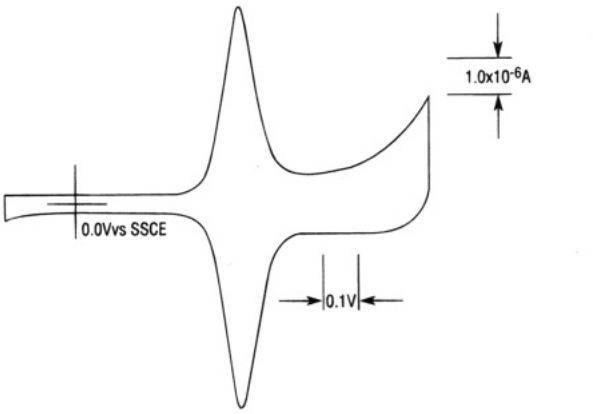



**Figure 3.** Cyclic voltammogram of an Au electrode coated with a mixed monolayer formed from $CH_3(CH_2)_7SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_8SH$; $N_T$ (total number of ferrocene moieties in this monolayer) = $0.55 \times 10^{-10}$ mol; $T$ = room temperature = 26.0 °C; scan rate = 0.1 V/s; $E^\circ$ = 0.469 V vs SSCE.

had a total (gold plus titanium) nominal thickness of 1.0 μm. The titanium underlayer causes a uniform orientation of the 111 facets of the gold microcrystallites comprising the vapor deposited gold layer.[4] These electrodes were cleaned in an argon-ion plasma before use.[25] Electrode area was 0.71 $cm^2$.

The self-assembled thiol monolayers were formed as described previously by Chidsey et al.[5] A clean electrode was placed in an ethanol solution containing a $1.0 \times 10^{-3}$ M total concentration of ferrocene-terminated thiol and diluent alkanethiol (the mole fraction of the ferrocene-terminated thiol was between 0.10 and 0.25) for approximately 16 h. The electrode is then taken out of this solution and rinsed in a succession of pure ethanol, *n*-hexane, ethanol, 95% ethanol/water, and water. This electrode is then attached to the ILIT cell (see Figure 2 and Figure 3 of reference 25). A major concern is that ferrocene moieties adsorbed at or close to monolayer defect sites could catalyze electron transfer to other ferrocene moities, thus enhancing the measured rate of electron transfer.[5] The sensitivity of the rate of electron transfer to the presence of monolayer defect sites acting as catalysts is expected to be greater for longer alkanethiol chains with their slower intrinsic rates of electron transfer. To establish whether or not this enhancement was occurring we immersed a monolayer with $n = 9$ (comprising approximately 10% of the tethered ferrocene) in an "exchange" solution consisting of $1.0 \times 10^{-3}$ M $CH_3(CH_2)_8SH$ in ethanol for approximately 2 weeks before performing the ILIT experiment on it. Ferrocene-terminated thiols which are associated with defect sites have been shown to be most facilely replaced by the unsubstituted alkanethiols.[3,5,7] Although there was some loss of tethered ferrocene from the surface, the electron-transfer rate constant obtained from the monolayer which had been placed in an "exchange" solution is the same as that obtained from those monolayers which had not been placed in an "exchange" solution. This suggests that if ferrocene moieties are close to defect sites in the original monolayer, they do not significantly enhance the rate of electron transfer in the present study.

The electrolyte for all experiments was 1.0 M $HClO_4$. Because of the limited solubility of $KClO_4$, a saturated sodium chloride calomel reference electrode (SSCE: $E_{SSCE} = E_{SCE} - 0.0052$ V; $E_{SCE}$ is the potential of the saturated (KCl) calomel electrode) was used instead of the usual saturated (potassium) calomel reference electrode. Cyclic voltammograms were performed on the thiol monolayers before each ILIT experiment. Figure 3 shows an example of such a cyclic voltammogram for $n = 8$. The capacitances of these monolayer films, $C_{film}$, were determined from the currents in the "double-layer region" of

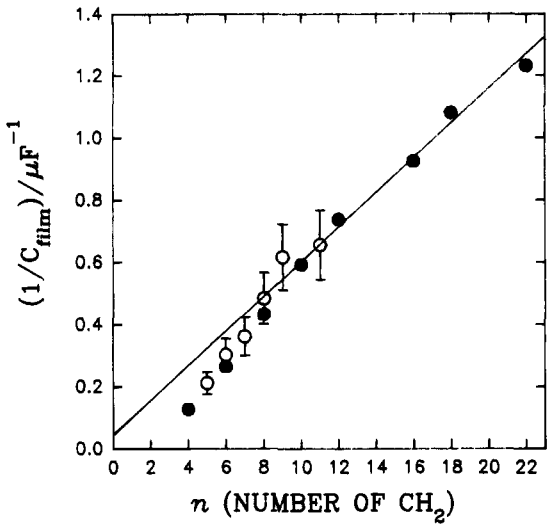



**Figure 4.** Plot of $1/C_{film}$ vs $n$. Open circles: capacitances determined in this work (see eq 8) for monolayers containing the ferrocene moiety. Filled circles: data for alkanethiol monolayers (no ferrocene) in contact with 1.0 M $HClO_4$, from ref 27. Line: least-squares fit of straight line for $n > 8$.

the cyclic voltammograms:

$$C_{film} = \frac{1}{2}\left|\frac{i_{E,anodic} - i_{E,cathodic}}{dE/dt}\right| \tag{8}$$

where $i_{E,anodic}$ and $i_{E,cathodic}$ are, respectively, anodic and cathodic charging currents measured at a potential ($E$) in the "double-layer region"; typically, $E = +0.1$ V versus SSCE and $|dE/dt|$ is the scan rate (=0.1 V/s). Porter et al.[27] have suggested the following simple relationship between $C_{film}$ and $n$:

$$C_{film} = \frac{\epsilon\epsilon_0}{d_{film}} = \frac{1}{a + bn} \tag{9}$$

where $\epsilon$ is the dielectric constant of the film, $\epsilon_0$ is the permittivity of free space, $a$ and $b$ are constants and $n$ is the number of methylene groups in the alkanethiol tether or diluent. A plot of $1/C_{film}$ vs $n$, for values of $C_{film}$ which we extract from cyclic voltammograms of alkanethiol tethered ferrocene films is shown in Figure 4. Our data are consistent with the results obtained by Porter et al.,[27] whose data for alkanethiol monolayers (containing no ferrocene moieties) in contact with 1.0 M $HClO_4$ is also shown in Figure 4. This agreement suggests that the structures (or lack thereof) of the films with or without ferrocene are comparable. The deviation from the straight line in Figure 4 for $n < 8$ could also indicate a change in the structure of the monolayer, a conclusion that is consistent with observations of other workers.[27] However, the measurements of Porter et al.[27] for the same films in 1 M NaF show no such deviation.

A curious feature of the cyclic voltammetry is the marked increase in the double layer charging current (Figure 3) when the ferrocene is fully oxidized (i.e., when $E_{app} > 0.650$ V versus SSCE). Chidsey and co-workers[28] have found that the capacitance of an oxidized ferrocene monolayer (for $n = 11$), deduced from impedance measurements, decreases as the frequency is increased beyond 100 Hz. The capacitance of the reduced film is independent of frequency over the same frequency range. The high-frequency capacitance of the oxidized film is comparable to the (high or low frequency) capacitance of the reduced film. A possible reason for this frequency dispersion is that the oxidized film is more permeable to ions and the resultant increase in the ion partition coefficient allows an ionic space charge to develop within the film. The resistance of the film to ion transport could affect the observed frequency dispersion. We presume that the same phenomena are operative in films for $n \leq 9$ which we investigate in this paper. Because the ILIT measurements reported here are completed in less than $10^{-5}$ s, we think it is valid to use the smaller value of $C_{film}$ (effectively, the high-frequency value) evaluated from the cyclic voltammogram at a potential where the ferrocene moieties are all reduced.

The number of ferrocene moieties (oxidized and reduced), $N_T$, in a monolayer is determined by integrating the rising portion of the anodic peak in the cyclic voltammogram (between 0.3 V vs SSCE and the peak potential), correcting the resulting integral for the integrated charging current, and multiplying the resulting value by $2/F$, where $F$ is the Faraday. We used this procedure to minimize corrections for the charging current baseline which increases rapidly at potentials postive of the $E°$. We have tried a number of protocols all of which give approximately the same results.

The change in the open-circuit potential ($\Delta V_t$) as a function of time is described by a master equation:[25]

$$\Delta V_t = A\Delta T_t^* + k_m B\int_0^t e^{-k_m(t-\tau)}\Delta T_\tau^*\, d\tau \tag{10}$$

$A\Delta T_t^*$ is potential dependent and is the initial potential response in the absence of any electron transfer. The measured rate constant of relaxation is $k_m$; $\Delta T_t^*$ is defined as[25]

$$\Delta T_t^* = \Delta T_t/\Delta T_{eq} \tag{11}$$

in which $\Delta T_t$ is the interfacial temperature change at time $t$ and $\Delta T_{eq}$ is the interfacial temperature perturbation that would be produced if all the absorbed heat were uniformly distributed in the electrode with no heat loss to either the dielectric ("backing") material or to the electrolyte solution; $B\Delta T_t^*$ is the amplitude of that portion of the response which results from the electron transfer (in the limit $t \rightarrow \infty$, the second term in eq 10 $\rightarrow B\Delta T_t^*$), where $B$ is given by

$$B = \frac{\Delta V_{redox} - A}{1 + \dfrac{RTC_{film}}{F^2 N_T}\left[\dfrac{(1+\omega)^2}{\omega}\right]} \tag{12}$$

where $\omega$ is the ratio of the surface coverages of the oxidized ($N_{ox}$) to the reduced ($N_{red}$) forms of ferrocene in the film and is obtained from the Nernst equation:

$$\omega = \frac{N_{ox}}{N_{red}} = \exp\left(\frac{F}{RT}(E_{app} - E°)\right) \tag{13}$$

$\Delta V_{redox}$ is the change in the equilibrium potential of the redox couple effected by the temperature perturbation and can be deduced directly from the Nernst equation:

$$\Delta V_{redox} = \Delta T_{eq}\frac{dE°}{dT} + \frac{\Delta T_{eq}}{T}(E_{app} - E°) \tag{14}$$

where $E_{app}$ is the potential of the electrode just prior to the perturbation.

The time dependence of the temperature change, $\Delta T_t$, has been discussed previously[25] and is a complicated, but analytic, function of the instrument response function (which includes the temporal shape of the laser pulse), the thickness of the metal film and the thermal properties (heat capacities and thermal

conductivities) of the backing, the metal, and the solution. The unknowns in eq 10 are $A$, $B$, and $k_m$. The least-squares optimization procedure used to fit the theory to experimental data and extract values of $A$, $B$, and $k_m$ has been described previously.[26]

For a surface attached redox couple, the definition of $k_m$ can be deduced from the Butler Volmer equation:[24]

$$k_m = \left[1 + \frac{RTC_{film}}{F^2}\left(\frac{1}{N_{ox}} + \frac{1}{N_{red}}\right)\right]\frac{F^2}{RTC_{film}} k_{\Gamma,s} N_{ox}^{1/2} N_{red}^{1/2} \quad (15)$$

where $C_{film}$ is the capacitance of the film (farads).[29] We assume that there is no significant diffuse double-layer potential on the solution side of the ferrocene moieties. Implicit in eq 15 is the presumption that the heterogeneous electron-transfer kinetics are measured at a potential close enough to $E°$ so that the Marcus expressions, eqs 3−6, are well approximated by the Butler−Volmer equation with a value of the transfer coefficient, α, equal to 0.5:

$$f_e = k_{\Gamma,s}\left(N_{ox} \exp\left[-\frac{1}{2}\frac{e}{k_B T}(E - E°)\right] - N_{red} \exp\left[\frac{1}{2}\frac{e}{k_B T}(E - E°)\right]\right) \quad (16)$$

where $f_e$ is the flux of electrons between the electrode and the redox moieties, and $k_{\Gamma,s}$ ($s^{-1}$) is the same standard rate constant as that defined in eq 6. Equation 15 can then be written as

$$k_m = \left[1 + \frac{RTC_{film}(1+\omega)^2}{\omega N_T F^2}\right]\frac{F^2 k_{\Gamma,s} N_T \omega^{1/2}}{RTC_{film}(1+\omega)} \quad (17)$$

Equation 17 offers a method for verifying that the observed relaxation exhibits the expected potential dependence, thereby verifying that it is a manifestation of electron transfer and also provides a method for computing $k_{\Gamma,s}$ from values of $k_m$ measured at different potentials.

All ILIT responses of the self-assembled ferrocene monolayers were measured as a function of applied potential ($E_{app}$). The applied potential was set with a specially designed potentiostat[25] whose potentiostating response can be "turned off" just before the laser pulse. Whatever current was required to sustain the extant potential just before "turn-off" is kept constant through the experiment.

The synthesis of the ferrocene-terminated thiols is described in ref 5. Aldrich alkanethiols were used as received. J. T. Baker Ultrex ultrapure $HClO_4$ was also used as received. As before, water was purified in a Labconco Water Prodigy.

## Results and Discussion

ILIT responses for $n = 5$ are shown in Figure 5. Figure 5a shows the response for a self-assembled monolayer formed from the ferrocene and thiol moieties. The transient was obtained at $E_{app} = 0.475$ V vs SSCE which is approximately the $E°$ for the system, which is the potential that effects the maximum value of $k_m$ (see eq 17). The solid line is the theoretical least-squares fit of eq 10 to the data giving $k_m = 7.8 \times 10^7$ $s^{-1}$; the dashed line is the predicted thermal response (i.e., when the open circuit potential tracks the interfacial temperature) in the absence of electron transfer. Figure 5b shows the response for the monolayer when $E_{app} = 0.250$ V vs SSCE $\cong E° - 0.225$ V. Under these conditions the value of $\omega$ (eq 13) becomes very small, the term $(1 + \omega)^2/\omega$ (eq 12) becomes very large and $B$ becomes very small. Thus the ILIT response is almost perfectly described by eq 10 with $B = 0$. The potential dependence of

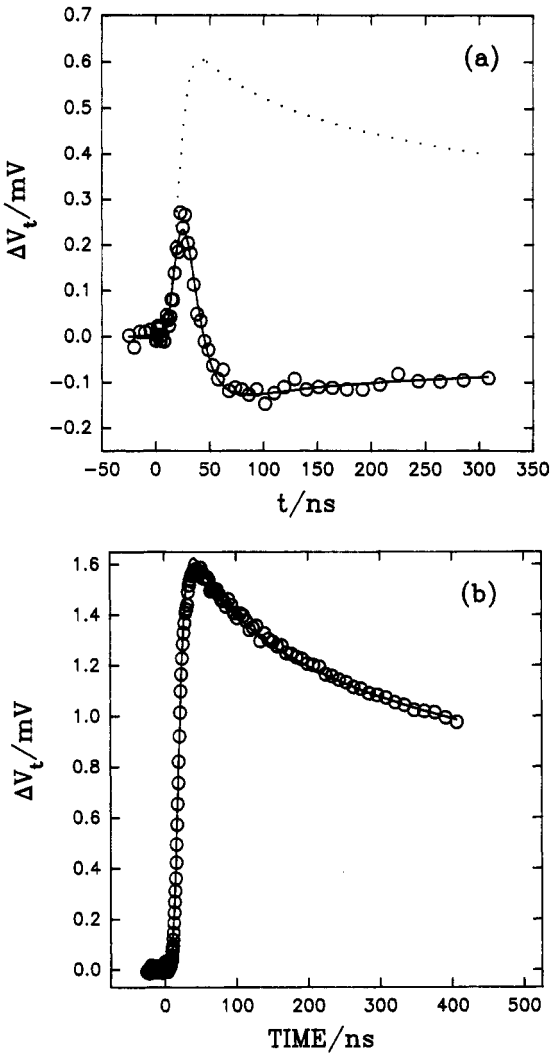



**Figure 5.** ILIT responses obtained from an Au electrode coated with a mixed monolayer formed from $CH_3(CH_2)_4SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_5SH$ with $N_T = 1.3 \times 10^{-10}$ mol; $T$ = room temperature = 31.0 °C, thickness of the electrode = $0.96 \times 10^{-4}$ cm. Open circles are experimental data. (a) $E_{app} = 0.475$ V vs SSCE; solid line is fit to eq 10, where $k_m = 7.8 \times 10^7$ $s^{-1}$, $A = 7.6 \times 10^{-4}$ V and $B = -9.2 \times 10^{-4}$ V. The dotted line represents the response which would be observed if there were no relaxation of the ILIT signal caused by electron transfer between the electrode and the redox couple. (b) $E_{app} = 0.250$ V vs SSCE; solid line is fit of this data to eq 10, for a purely thermal response ($k_m = 0$; $B = 0$) with $A = 2 \times 10^{-3}$ V.

$k_m$ for this film is shown in Figure 6. The good fit to eq 17 supports the contention that the relaxations are caused by electron transfer.

Figure 7 shows the much slower response for a self-assembled monolayer comprising the tethered ferrocene diluted with the corresponding alkanethiol for $n = 9$. The solid line is the least-squares optimized fit for $k_m = 6.9 \times 10^5$ $s^{-1}$.

Table 1 contains the values of $k_{\Gamma,s}$ (determined from eq 17 for values of $k_m$ measured at different potentials) for various values of $N_T$, $C_{film}$, and $n$ at $T = 25$ °C. Because eq 17 contains the ratio $N_T/C_{film}$, the electrode roughness factor cancels out in the computation of $k_{\Gamma,s}$ from $k_m$; note that for a given value of $n$, $k_{\Gamma,s}$ is found to be independent of $N_T/C_{film}$. Figure 8 shows plots of $\ln[k_{\Gamma,s}]$ versus $n$ at five temperatures. The slopes and intercepts of these plots ($\beta_n$ and $\ln[k_{\Gamma,s,n=0}]$, see eq 1) are given in Table 2. There is a slight decrease in $\beta_n$ with increasing $T$ (see Table 2) although the range of values is comparable to the experimental error. Figure 9 shows the same data, plotted as a set of Arrhenius plots, $\ln[k_{\Gamma,s}/T^{1/2}]$ versus $1/T$, for $n = 5-9$.

A technical aside is required here. Evaluation of $k_{\Gamma,s}$ at a

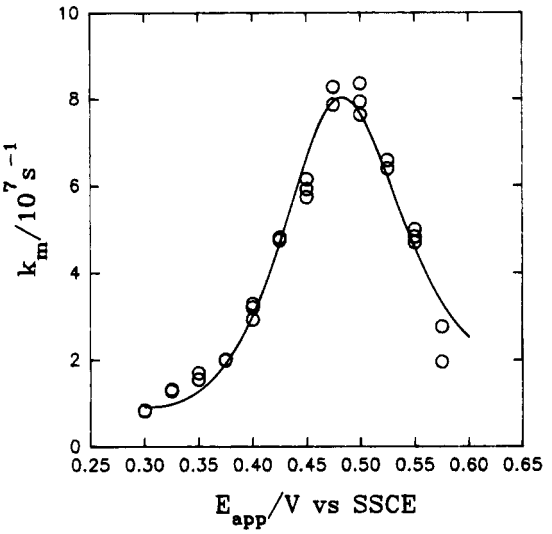



**Figure 6.** Plot of $k_m$ versus $E_{app}$ (same monolayer as in Figure 5). The solid line is the least-squares fit to eq 17 giving $k_{\Gamma,s} = 1.6 \times 10^6\ s^{-1}$.

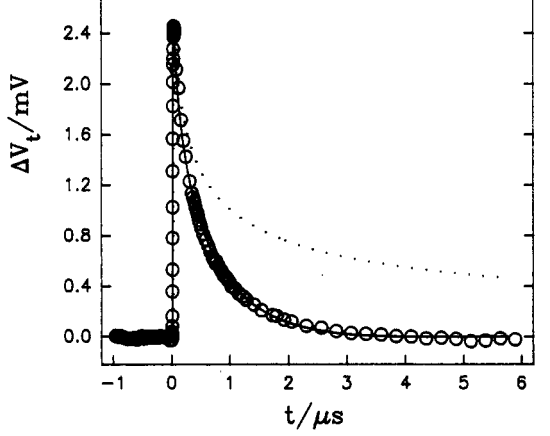



**Figure 7.** ILIT response obtained from an Au electrode coated with a mixed monolayer formed from $CH_3(CH_2)_8SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_9SH$ where $E_{app} = E° = 0.475$ V vs SSCE, $N_T = 0.52 \times 10^{-10}$ mol; $T$ = room temperature = 29.0 °C, thickness of the electrode = $0.78 \times 10^{-4}$ cm. Open circles are experimental data. Solid line is fit of this data to eq 10 where $k_m = 6.9 \times 10^5\ s^{-1}$, $A = 3.29 \times 10^{-3}$ V, and $B = -2.79 \times 10^{-3}$ V. The dotted line represents the response which would be observed if there were no relaxation of the ILIT signal caused by electron transfer between the electrode and the redox couple.

given temperature, $T$, requires the value of $C_{film}$ at $T$ (see eq 17). The change in $C_{film}$ as a function of $T$ is too small to measure accurately using cyclic voltammetry. However, we have a much more sensitive method to estimate $dC_{film}/dT$: the initial response of the double layer to the temperature perturbation (i.e., the quantity $A$, see eq 10). The value of $A$ varies with applied potential, $E_{app}$, according to[24]

$$\frac{A}{\Delta T_{eq}} = -\frac{d\ \ln(C_{film})}{dT}(E_{app} - E_{pzc}) + \frac{dE_{pzc}}{dT} \qquad (18)$$

where $E_{pzc}$ is the potential of zero charge. The slope of a plot of $A/\Delta T_{eq}$ versus $E_{app}$, will be $-d\ \ln(C_{film})/dT$ if $dE_{pzc}/dT$ is constant. Figure 10 shows such a plot for $n = 6$. It is linear suggesting that $d\ \ln(C_{film})/dT$ is independent of potential over the range of interest. The slopes of similar plots for different values of $n$ are approximately independent of $n$ and temperature.

**TABLE 1: Electron-Transfer Rate Constants ($k_{\Gamma,s}$), $C_{film}$, and $N_T$ and the Quantity $N_T/C_{film}$ at $T = 25$ °C**

| $n$ | $C_{film}$/Farad[a] | $N_T$/mol[a] | $(N_T/C_{film})$ (mol/Farad)[b] | $k_{\Gamma,s}$ /s$^{-1}$ |
|---|---|---|---|---|
| 5 | $5.8 \times 10^{-6}$ | $1.3 \times 10^{-10}$ | $2.2 \times 10^{-5}$ | $1.6 \times 10^{6}$[d] |
| | $6.2 \times 10^{-6}$ | $1.2 \times 10^{-10}$ | $1.9 \times 10^{-5}$ | $1.6 \times 10^{6}$[d] |
| | $3.7 \times 10^{-6}$ | $0.35 \times 10^{-10}$ | $0.95 \times 10^{-5}$ | $1.6 \times 10^{6}$[d] |
| | $3.7 \times 10^{-6}$ | $0.28 \times 10^{-10}$ | $0.76 \times 10^{-5}$ | $1.6 \times 10^{6}$ |
| | $5.2 \times 10^{-6}$ | $0.33 \times 10^{-10}$ | $0.63 \times 10^{-5}$ | $1.5 \times 10^{6}$ |
| 6 | $3.4 \times 10^{-6}$ | $0.83 \times 10^{-10}$ | $2.4 \times 10^{-5}$ | $2.4 \times 10^{5}$ |
| | $2.1 \times 10^{-6}$ | $0.52 \times 10^{-10}$ | $2.5 \times 10^{-5}$ | $2.5 \times 10^{5}$[d] |
| | $2.9 \times 10^{-6}$ | $0.32 \times 10^{-10}$ | $1.1 \times 10^{-5}$ | $2.3 \times 10^{5}$ |
| 7 | $2.9 \times 10^{-6}$ | $0.94 \times 10^{-10}$ | $3.3 \times 10^{-5}$ | $1.6 \times 10^{5}$[d] |
| | $2.7 \times 10^{-6}$ | $0.50 \times 10^{-10}$ | $1.9 \times 10^{-5}$ | $1.7 \times 10^{5}$ |
| 8 | $2.1 \times 10^{-6}$ | $0.55 \times 10^{-10}$ | $2.6 \times 10^{-5}$ | $3.3 \times 10^{4}$[d] |
| | $2.4 \times 10^{-6}$ | $0.55 \times 10^{-10}$ | $2.3 \times 10^{-5}$ | $3.5 \times 10^{4}$ |
| | $1.8 \times 10^{-6}$ | $0.30 \times 10^{-10}$ | $1.7 \times 10^{-5}$ | $3.3 \times 10^{4}$ |
| 9 | $1.6 \times 10^{-6}$ | $0.52 \times 10^{-10}$ | $3.3 \times 10^{-5}$ | $1.0 \times 10^{4}$[d] |
| | $1.5 \times 10^{-6}$ | $0.22 \times 10^{-10}$ | $1.5 \times 10^{-5}$ | $1.1 \times 10^{4}$ |
| | $1.7 \times 10^{-6}$ | $0.17 \times 10^{-10}$ | $1.0 \times 10^{-5}$[c] | $1.1 \times 10^{4}$[d] |

[a] To convert to Farad/cm$^2$ or mol/cm$^2$ divide values by electrode area, 0.71 cm$^2$. [b] This is the actual term used in the calculations of $k_{\Gamma,s}$ from $k_m$ (see eq 17). [c] This monolayer spent approximately 2 weeks in an "exchange solution" consisting of $1.0 \times 10^{-3}$ M $CH_3(CH_2)_8SH$ in ethanol. [d] Rate constants measured at room temperature (29.5 ± 1.5 °C) and extrapolated to 25 °C using Arrhenius plots.

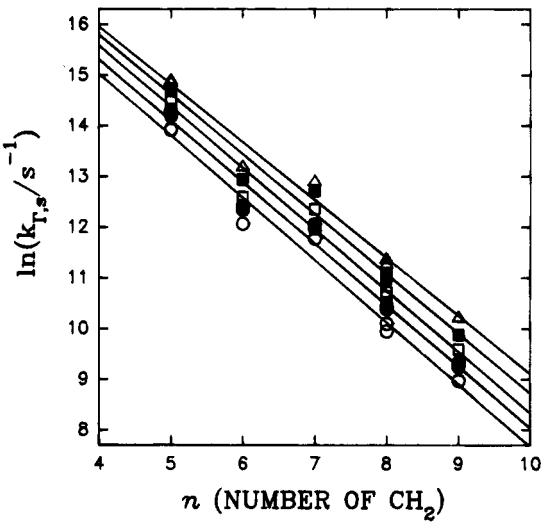



**Figure 8.** Plots of $k_{\Gamma,s}$ versus $n$ at five different temperatures. Open circles: $T = 14.9$ °C. Filled circles: $T = 25.0$ °C. Open squares: $T = 35.0$ °C. Filled squares: $T = 44.8$ °C. Open triangles: $T = 55.2$ °C. The slopes, $\beta_n$, of these plots, given in Table 2, were obtained by a linear least-squares fit to the data at each temperature.

**TABLE 2: Values of $\beta_n$ and $\ln[k_{\Gamma,s,n=0}]$ from the Plots in Figure 8**

| $T$ (°C) | $\beta_n$ | $\ln[k_{\Gamma,s,n=0}]$ |
|---|---|---|
| 14.9 | 1.23 ± 0.08 | 19.9 ± 0.6 |
| 25.0 | 1.21 ± 0.05 | 20.2 ± 0.3 |
| 35.0 | 1.21 ± 0.08 | 20.4 ± 0.5 |
| 44.8 | 1.18 ± 0.08 | 20.5 ± 0.6 |
| 55.2 | 1.14 ± 0.07 | 20.5 ± 0.5 |

The average value of $d\ \ln(C_{film})/dT$ is $-(1.27 \pm 0.25) \times 10^{-3}$ K$^{-1}$. The value of $C_{film}(T)$ is then well approximated by

$$C_{film}(T) = C_{film,T=303K}(1.385 - (0.00127K^{-1})T) \qquad (19)$$

Figure 11 shows a plot of $\ln[k_{\Gamma,s}]$ at 25 °C (open circles) replotted vs $n$ and compared with the value of $\ln[k_{\Gamma,s}]$ for $n = 16$ obtained using the chronoamperometric method[3] (filled circle). The chronoamperometric value falls very near the line that fits the ILIT data ($\beta_n = 1.21 \pm 0.05$; see Table 2), providing a check on the validity of both methods. The distance

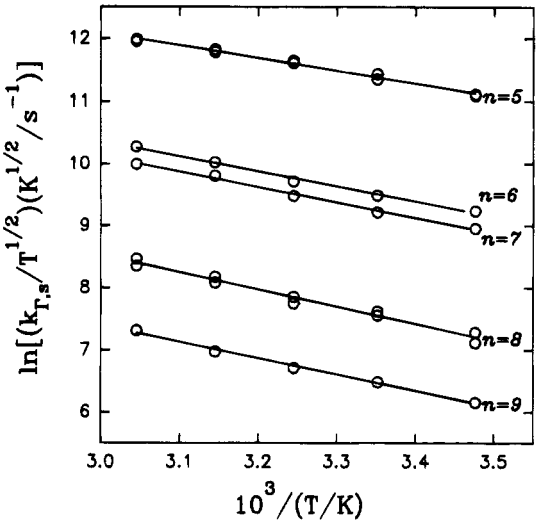



**Figure 9.** Arrhenius plots: semilogarithmic plots of $k_{\Gamma,s}/T^{1/2}$ vs $1/T$ (for $n = 5-9$) where the values of $k_{\Gamma,s}$ are obtained from mixed monolayers formed from $CH_3(CH_2)_{n-1}SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_nSH$. When there is more than one point at a particular temperature, this means that $k_{\Gamma,s}$ was determined from more than one monolayer. The slope for each of these lines and the corresponding values of $\lambda$ calculated from eq 7 are given in Table 3.

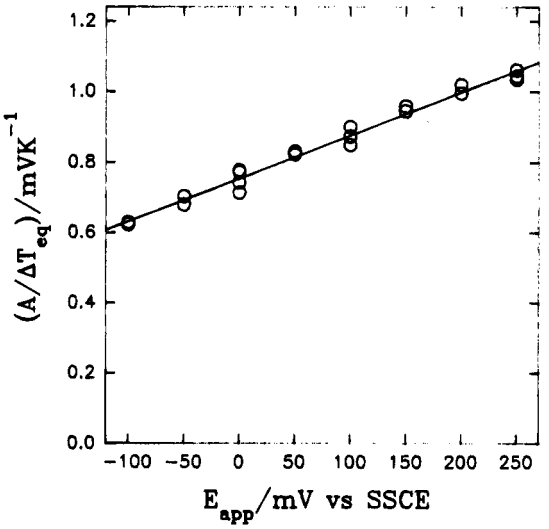



**Figure 10.** Plot of $A/\Delta T_{eq}$ versus $E_{app}$ for a mixed monolayer formed from $CH_3(CH_2)_5SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CO_2(CH_2)_6SH$ where $N_T = 0.52 \times 10^{-10}$ mol and $T = 29.5$ °C. The slope, $-d\ \ln[C_{film}]/dT$, and the intercept are $(1.22 \pm 0.04) \times 10^{-3}$ $K^{-1}$ and $(0.754 \pm 0.005) \times 10^{-3}$ V/K, respectively.

dependence for the interval $n = 7-18$ has been determined more recently by chronoamperometry ($\beta_n = 1.11 \pm 0.05$).[7] We are exploring possible explanations for the small but significant descrepancy between the values of $\beta_n$ obtained by the two methods.

We can also offer a simplistic analysis which suggests the values we are measuring for $k_{\Gamma,s}$ are consistent with reported values of $k_s > 1$ cm/s at 25 °C[30] for the heterogeneous electron-transfer rate constant between an electrode and (untethered) ferrocene moieties in nonaqueous media. If we assume that the plane of closest approach for the ferrocene moiety in this reaction is approximately 2 Å (i.e., the distance between the electrode and the surface of the redox moiety), this approximately corresponds to $n = 0$ because of the intervening Au−S bond and ester group involved in the ferrocene tether, which would locate the edge of a ferrocene moiety at about 2 Å and thus would produce a value of $k_{\Gamma,s} \sim 6 \times 10^8$ $s^{-1}$ at 25 °C. If we assume that the distance dependence of the electron transfer rate for species in bulk solution is also exponential with

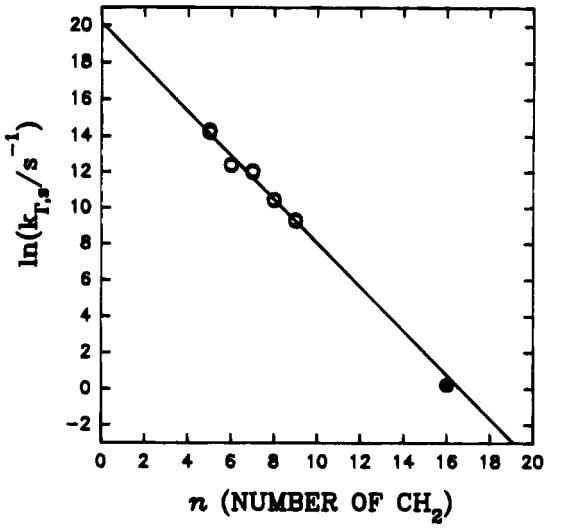



**Figure 11.** Plot of $\ln[k_{\Gamma,s}]$ vs $n$ at 25 °C. ILIT rate constants (○) are from Table 1. The value for $n = 16$ (●) from ref 3 was obtained by chronoamperometry. The solid line is a fit to the ILIT data giving $\beta_n = 1.21 \pm 0.05$ and $k_{\Gamma,s,n=0} = (5.7 \pm 2) \times 10^8$ $s^{-1}$ (see eq 1).

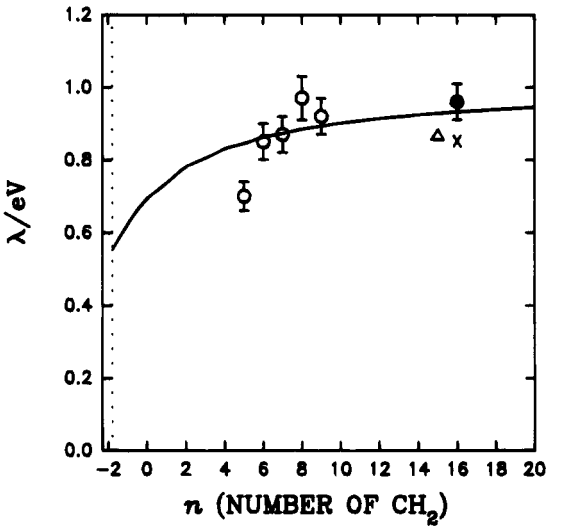



**Figure 12.** Plot of the reorganization energy, $\lambda$, vs $n$: (○) values of $\lambda$ computed from slopes of Arrhenius plots constructed from values of $k_{\Gamma,s}$ evaluated from ILIT data for $n = 5-9$ (see eq 7); (×) value of $\lambda$ determined by chronoamperometric measurements at high overpotential for $n = 16$;[3] (●) value of $\lambda$ determined from an Arrhenius plot of the data obtained by chronoamperometry at $E = E°$ and $n = 16$;[3] (△) value of $\lambda$ determined from cyclic voltammetry (effectively a high overpotential method) for *N*-(15-mercaptopentadecyl)ferrocenecarboxamide.[15] The solid line is computed from the theory in ref 22; dielectric constants of linear polyethylene[39] were used for the alkanethiol monolayers, the thicknesses of the monolayers were determined from eqs 21 and 22, and the radius of the ferrocene moiety was assumed to be $3.6 \times 10^{-8}$ cm.[15] The vertical dotted line indicates the position of the gold surface. (It is at a negative value of $n$ because $n$ indicates the number of methylene carbons in the thiol tether, which is ultimately attached to the gold electrode through the terminal sulfur atom.)

$\beta_r \sim 1$ Å$^{-1} = 1 \times 10^8$ cm$^{-1}$, then the integrated rate from the plane of closest approach (PCA) to infinity would be

$$k_s \sim 6 \times 10^8 \int_{x_{PCA}}^{\infty} e^{-\beta_r x}\, dx = \frac{6 \times 10^8}{\beta_r} = 6 \text{ cm/s} \quad (20)$$

An estimate of 6 cm/s for $k_s$ is quite reasonable.[30]

Using eq 7, the value of the reorganization energy, $\lambda$, is deduced from the slopes of the Arrhenius plots in Figure 9. A plot of $\lambda$ as a function of $n$ is shown in Figure 12. Values of $\lambda$ are deduced from slopes of Arrhenius plots constructed from values of $k_{\Gamma,s}$ measured using ILIT (○) for $n = 5-9$. (×)

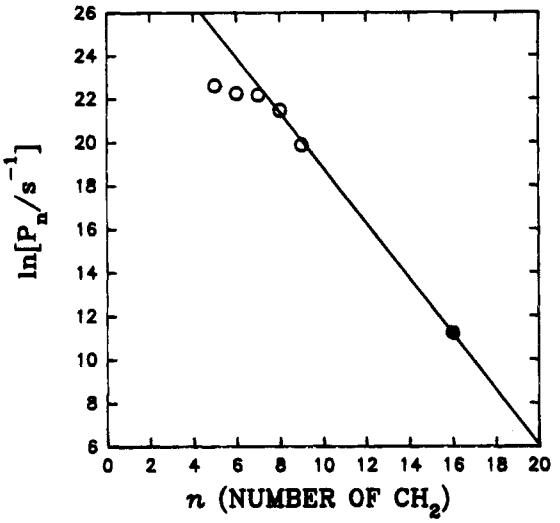



**Figure 13.** Plot of $\ln[P_n]$ vs $n$ for all temperatures (error bars are smaller than points). $P_n$ is defined in eq 5 and computed from eq 6. The line is a linear fit for $n \geq 8$ with slope $= -1.27 \pm 0.1$ and intercept $= 30.6 \pm 0.3$.

indicates the value of $\lambda$ determined by chronoamperometric measurements at high overpotential for $n = 16$;[3] (●) indicates the value of $\lambda$ determined from an Arrhenius plot of the data obtained by chronoamperometry at $E = E°$ and $n = 16$;[3] (△) indicates the value of $\lambda$ determined from cyclic voltammetry (effectively a high-overpotential method) for *N*-(15-mercaptopentadecyl) ferrocenecarboxamide.[15] For $n \geq 8$, values of $\lambda$ determined by all methods are in reasonable agreement with a value of ~0.9 eV. We suggest that the decrease in the value of $\lambda$ with decreasing $n$ for $n = 8$, 7, 6, and 5 is significant. There appears to be a correlation between the corresponding values of $k_{\Gamma,s}$ and $\lambda$. The values of $P_n$ are also determined from the Arrhenius plots. A plot of $P_n$ as a function of $n$ is shown in Figure 13. This is a universal plot showing the data obtained at all temperatures. The scatter in the data for a given value of $n$ is less than the size of the point. The trends in the observed $P_n$ values may be compared with the results of a number of theoretical studies of intramolecular electronic coupling in radical cation or anion systems based on trans-staggered alkane spacer chains containing 2−19 methylene units with various terminal donor/acceptor groups, including $-NH_2$, $-CH_2$, and vinyl.[31−35] The overall variation of donor−acceptor coupling with $n$ depends upon the charge and chemical type of the donor−acceptor groups. For $n \geq 3$ the coupling always decreases monotonically and in a roughly exponential manner with increasing $n$; mean values of $\beta_r$ ranging generally from ~0.5 to ~1.0 $Å^{-1}$ have been calculated. The range in $\beta_r$ reflects, to a large extent, the variation in the gap between the donor−acceptor and spacer energy levels.[36] In the case of $-CH_2$ donor−acceptor groups a flattening or slight turnover of $P_n$ has been observed as one proceeds from $n = 3$ to $n = 2$.[32,33] The experimental value of $\beta_n$ (Table 2) corresponds approximately to a value of $\beta_r \simeq 1$ $Å^{-1}$ and approximates the upper limit for $\beta$. The apparent flattening in $P_n$ which we observe for the heterogeneous electron transfer between the gold electrode and the ferrocene moities when $n \leq 7$ (Figure 13) is occurring at much larger than theoretically predicted distances.

Is there some theoretical basis for a decrease in the value of $\lambda$ at smaller values of $n$? Liu and Newton[22] have predicted a decrease in the reorganization energy with decreasing film thickness. The results of this continuum-based analysis depend upon film thickness, static and optical dielectric constants of the film and aqueous phase, the radius of the ferrocene moiety, and its location (assumed to lie just outside the film/solution interface), and can be compared with our experimental plot of $\lambda$ vs $n$ (solid line, Figure 12). The slight irregularity in the theoretical relationship arises because the distance between the gold surface and the redox moiety does not increase uniformly with $n$.[22] Different algorithms relate the distances for odd and even values of $n$:

$$d_{odd} = [1.12(n_{odd} - 1) + 2.9]\ Å \quad (21)$$

$$d_{even} = [1.12n_{even} + 2.0]\ Å \quad (22)$$

where $d_{odd}$ or $d_{even}$ is the distance from the electrode surface to the outer edge of the terminal methyl group of the diluent alkanethiol. These equations evolve from a model for the structure of these monolayers in which the alkyl chain is all *trans* with its axis canted 27° from the surface normal to the electrode,[27] and all carbon atoms in the alkyl chain are in the same plane and that plane is rotated by 55° from the plane defined by the chain axis and the surface normal.[37] The model also assumes that, because of the intervening sulfur atom, the first carbon atom is 2 Å away from the gold surface.

The theory (solid line, Figure 12) does not predict as great a decrease in $\lambda$ with decreasing $n$ as is experimentally observed for $n \leq 8$. There are other factors which may be playing a role. Several authors have reported a decrease in the structural order of alkanethiol films with decreasing $n$,[37] a phenomenon that may also be reflected in the capacitive behavior which we discussed earlier.[27] Perhaps factors other than the reorganization energy are affecting the slopes of the Arrhenius plots at smaller values of $n$.[26] Nevertheless there appears to be a definite correlation between $k_{\Gamma,s}$ and $\lambda$: the relationship between $k_{\Gamma,s}$ and $n$ for $n \leq 8$ is not perfectly smooth (Figure 8; note that the irregularity occurs at all temperatures). An irregularity also occurs in the relationship between $\lambda$ (and therefore the slope of the Arrhenius plot) and $n$ (Figure 12). However, the smooth relationship between $P_n$ and $n$ (Figure 13) suggests that the values of $k_{\Gamma,s}$ and $\lambda$ (for $n \leq 8$) are coupled.

## Conclusions

An interpretation of the data we have presented suggests that the dependence of the prefactor $P_n$ (eq 6) on $n$, the number of carbons between the ferrocene moiety and the sulfur, is not strictly exponential and that the slope of the plot of $\ln[P_n]$ vs $n$ flattens when $n \leq 7$ (see Figure 13). A corollary conclusion is that when $n < 8$ the continued linear relationship between $\ln[k_{\Gamma,s}]$ and $n$ for the plots in Figures 8 and 11 reflects a fortuitous combination of decreasing values of $\lambda$ (which would tend to increase the value of $k_{\Gamma,s}$) and a flattening of the slope of $\ln[P_n]$ vs $n$ (which tends to decrease the value of $k_{\Gamma,s}$). However, as noted earlier, theoretical calculations do not predict such a flattening, although in some cases, for very short spacer lengths ($n = 2-3$), some flattening, or even nonmonotonic behavior is observed.[32,33]

The experimental value of $\beta_n = 1.21 \pm 0.05$ (Figure 11) for our ferrocene system is similar to the value of $\beta_n = 1.06 \pm 0.04$ measured by Finklea for $PyRu(NH_3)_5$ moieties with alkanethiol tethers and matching length diluent alkanethiols[8] and to the values of $\beta_n$ measured for self-assembled thiol films (on gold electrodes) with redox moieties in solution: $1.08 \pm 0.2$ for aqueous $Fe(CN)_6^{3-/4-}$, $Ru(NH_3)_6^{3+/2+}$, and $Fe^{3+/2+}$ with $\omega$-hydroxythiol films ($n = 6-22$);[20] and $1.02 \pm 0.2$ for aqueous $Fe(CN)_6^{3-/4-}$ and $Fe^{3+/2+}$ with alkanethiol films ($n = 12-18$).[38] The corresponding value of $\beta_r \simeq 1$ $Å^{-1}$ is also similar to values inferred from experimental kinetic data for saturated spacers (e.g., see data cited in refs 11 and 33) and lies within the upper range of estimates (~0.5−1.0 $Å^{-1}$) obtained from the model calculations[31−35] discussed above. We emphasize that the

**TABLE 3: Slopes (d ln[$k_{\Gamma,s}/T^{1/2}$]/d[1/$T$]) of the Lines in Figure 9 and Reorganization Energies ($\lambda$) Determined from These Slopes Using Eq 7**

| $n$ | d ln[$k_{\Gamma,s}/T^{1/2}$]/d[1/$T$]/K | $\lambda$/eV |
|---|---|---|
| 5 | $-(2.0 \pm 0.1) \times 10^3$ | $0.70 \pm 0.04$ |
| 6 | $-(2.4 \pm 0.1) \times 10^3$ | $0.84 \pm 0.05$ |
| 7 | $-(2.5 \pm 0.2) \times 10^3$ | $0.86 \pm 0.05$ |
| 8 | $-(2.8 \pm 0.2) \times 10^3$ | $0.96 \pm 0.06$ |
| 9 | $-(2.6 \pm 0.1) \times 10^3$ | $0.91 \pm 0.05$ |

calculated values of $\beta$ reflect the energy gaps[36] for the particular model systems dealt with (see especially ref 34, which offers examples of very small gaps and hence unusually small values of $\beta$). Estimates of the corresponding gaps relevant to the experimental systems are not readily available, and thus further detailed comparison of calculated values of $\beta$ and experimental values of $\beta_r$ is not warranted at the present time. However, the experimental values of $\beta_r$ may be considered as upper limits to the true electronic decay coefficient, $\beta$, since correction for the $n$-dependence of $\lambda$ would yield some reduction in the estimate of the magnitude of $\beta$.[14,22] Such an effect would be expected to be important only for $n \leq 6$, based on the estimates for $\lambda$ given in Table 3.

The ambiguity of the structure of these films for $n \leq 9$ complicates the interpretation of our data. It is reassuring that the values of $\lambda$ deduced from the slopes of Arrhenius plots for $n \geq 8$ agree well with the value of 0.85 eV which Chidsey deduced directly from large overpotential chronoamperometric experments.[7] Interestingly, Weber and Creager, using cyclic voltammetry, measure $\lambda = 0.87$ eV for monolayers formed from $HOCH_2(CH_2)_{15}SH$ and $(\eta^5\text{-}C_5H_5)Fe(\eta^5\text{-}C_5H_4)CONH(CH_2)_{15}SH$.[15] The decrease in $\lambda$ suggested by our ILIT data for $n < 8$ cannot be completely explained by the theory of Liu and Newton.[22] Unfortunately, when $n < 8$, $\lambda$ cannot be measured using a direct large overpotential method[3,7] (the electron-transfer rates are too fast); consequently, the relationship between $\lambda$ and the slope of the Arrhenius plot (eq 7) cannot be verified as was done for $n = 16$.[3] It is also possible that the structure of the films for $n < 8$ is temperature dependent and that could further complicate the relationship between $\lambda$ and the slope of the Arrhenius plot. A more detailed picture of the structures of these mixed films would be invaluable.

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## References and Notes

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